

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000782		
	Version:	9.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	24-Apr-2018
Title:	SM DG57. LIS connection			

DG-57 Service Manual

LIS CONNECTION

Procedure description
-LIS connection. -Procedure: describes how to connect the system to a LIS computer. -HostDemo software. -LIS communication protocol. In this section LIS communication protocol is detailed. <i>Note: Document REGD-0000782_v9 replaces document REGD-0000782_v8</i>

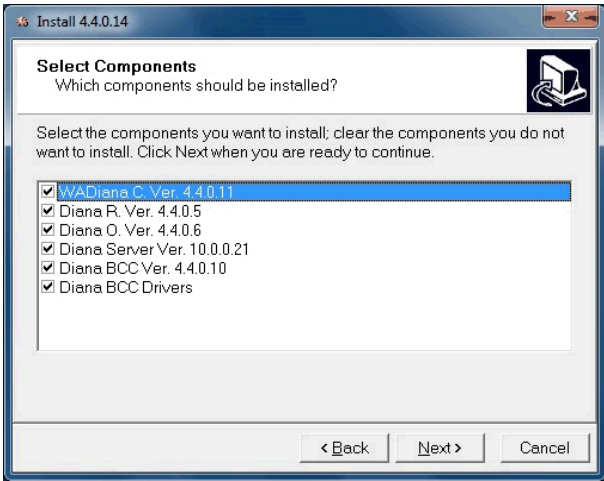
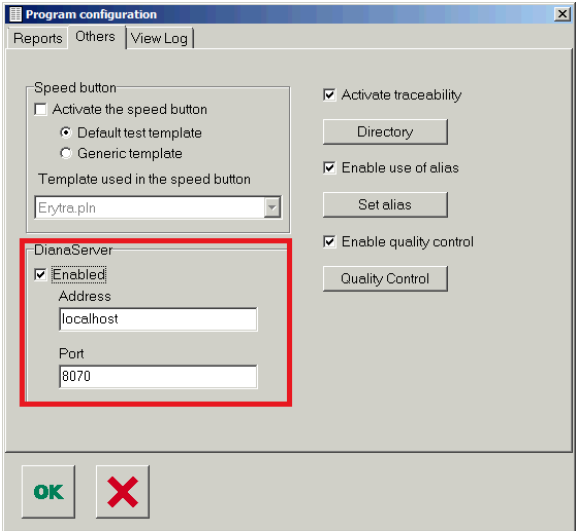
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LIS CONNECTION.

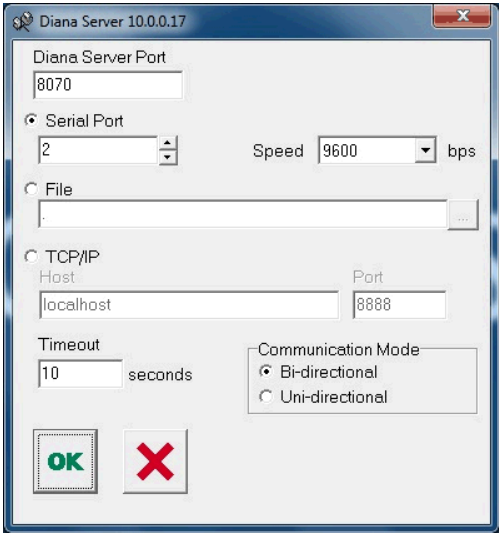
Process	Expected Results
The connection of WADiana to the <i>LIS</i> is established by means of <i>Diana System</i> (DianaServer program) and Diana System names (tests, microtubes,...) will be used. <i>Diana System</i> to <i>LIS</i> communication has been implemented in accordance with the CLSI LIS01-A2 and CLSI LIS02-A2 standards (formerly ASTM E1381-02 and ASTM E1394-97, respectively). CLSI LIS02-A2 standard establishes a format for the messages that must be sent from the LIS to the instrument and vice versa, but does not specify the way in which they must be sent. CLSI LIS01-A2 standard describes the way in which messages should be physically sent to the LIS, but does not specify a format. <i>WADiana</i> can communicate with the <i>LIS</i> in 3 ways: • Following <i>CLSI LIS01-A2</i> standard, where the serial channels of the computers are used. • By writing the desired information in files on local or remote disk units that can be accessed by both, <i>LIS</i> and <i>WADiana</i>. • By using TCP/IP protocol in a network domain. DianaServer will temporary store the requests sent by the LIS in <i>LisTests.def</i> file. When a result has to be transmitted to the LIS, the transmission can follow two different modes: • bi-directional transmission: The existing record in the database is updated • uni-directional/re-send transmission: a new record is created in the database Once the transmission has been completed successfully, DianaServer deletes the transmitted record.	---

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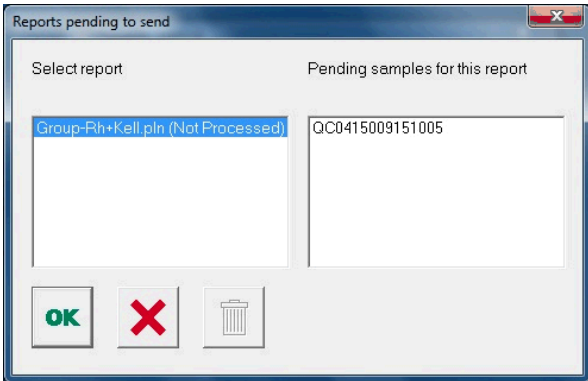
PROCEDURE: LIS connection.

Process	Expected Results
1. Make sure that DianaServer software has been properly installed in the computer that will be connected to the LIS system (see REGD-0000777).  Picture 1 2. Enable the use of DianaServer program by ticking the corresponding checkbox in Diana O. program (Listas.exe) located in <i>installation path</i>\SistemaDiana → <i>Program configuration</i> → <i>Others</i> tab (disabled by default). As the communication between DianaServer and WADiana C. software uses IP protocol, the <i>IP Address</i> and <i>Port</i> fields of the PC where DianaServer is going to be used should also be configured in this window (default values are <i>localhost</i> and <i>8070</i>, respectively).  Picture 2 3. Run DianaServer program from Diana System programs menu in Windows. Once executed, an icon appears in the bottom-right edge of the Windows taskbar (depending on your password configuration it will ask for a user – password pair to validate the permissions).	Software installed and usable. DianaServer is enabled and communication configuration is properly set. The following icon appears in Windows taskbar: 

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Process	Expected Results
4. Right clicking with the mouse brings up a context menu with the options <i>Configuration</i> and <i>Close</i>. Clicking on the <i>Configuration</i> option opens the following window:  Picture 3 - The DianaServer port to be used should be indicated in the <i>Diana Server Port</i> field on top of the window, and must match with the <i>Port number</i> indicated in Diana O. configuration (default value is 8070). - If <i>Serial Port</i> is selected the communication will use a physical port (number 2 /COM2 by default). Transmission speed can also be configured and must match the speed set by the LIS system. -If <i>File</i> is selected, the communication will use a shared directory in order to interchange data. Both computers have to be configured so that the same shared directory is used. Note: The exchange directory shall be unique. That is, if several systems should be connected to host and more than one DianaServer is used, each DianaServer must be configured with a different exchange directory. File exchange mode is as follows: When WADiana has to do a request/send a message to the LIS, DianaServer writes the corresponding commands (following the ASTM E 1394 91 standard) in a file named <i>TX_TMP.DAT</i> in the configured exchange directory. Once the complete message is written, this file is renamed to <i>OUT.DAT</i>. LIS shall read this <i>OUT.DAT</i> file and then delete it (in order to indicate that it has been read) within the allowed time set in <i>Timeout</i> parameter (default time is 10 seconds). If the file has not been deleted within the allowed timeout, an error is generated. LIS shall write the answer message in a file named <i>IN.DAT</i> located in the configured exchange directory. DianaServer will read this message and delete it to inform the LIS that the message has been read.	The configuration will be saved.

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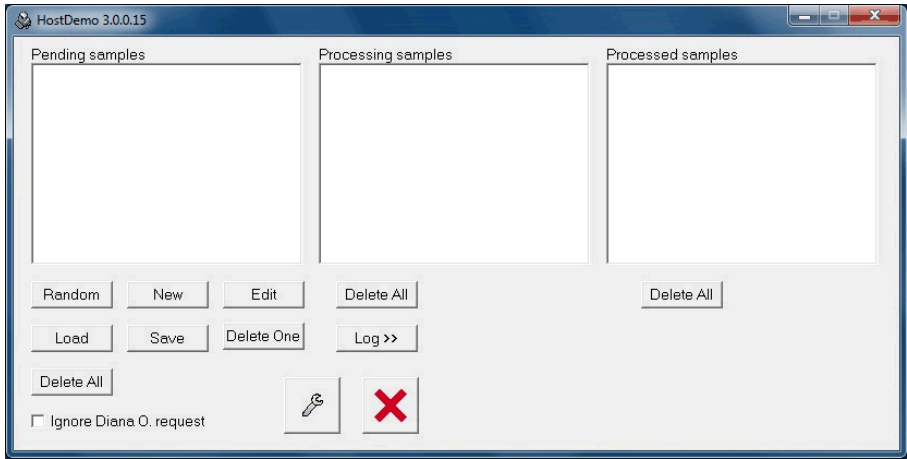
Process	Expected Results
- If <i>TCP/IP</i> is selected, the communication will use the TCP/IP protocol when exchanging information with the host. The corresponding IP address and port must be indicated. - The <i>Timeout</i> setting indicates the time in seconds that the DianaServer should wait before receiving an answer once a message has been sent to the LIS. - <i>Communication Mode</i> allows setting up communication between DianaServer and LIS: in uni-directional case DianaServer can only send test results to LIS, while in bi-directional case it can also receive programmed worksheets. Note that the LIS cannot start a communication with WADiana. When the desired parameters have been configured, click on <i>OK</i> button. 5. Now start Diana O or WADiana software to communicate with LIS computer. Windows Start -> Programs -> Sistema Diana -> WADiana C./ Diana O. If bidirectional mode has been selected, Inside WADiana C 'Sheet' tab it appears the LIS button called <i>Host</i>. The purpose of this button is receiving from LIS computer all tests to run to every sample. If bidirectional mode has been selected, Inside Diana O main windows it also appears a new 'LIS' button :  Picture 4 The purpose of this button is to send the results obtained in the processed tests that have been ordered by the LIS for specific samples requested by LIS to the Diana O Software. First Diana O. asks the LIS whether any specific sample results need to be sent. Once the first connection is completed, Diana O. reads all data from the local database, and then it shows the corresponding information in a new window (see below).  Picture 5	The communication will take place.

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Process	Expected Results
Samples are ordered in different reports: Each report is identified by the name of the template that will be used to interpret the results of the processed samples. The report is a Results by sample listing. The Diana O Software makes a distinction between reports generated as a result of an order to perform a test and an order to send results. Results of the latter type are displayed with the notation (Not processed) next to the report name. As long as DianaServer is running, when listing results by sample in Diana O (both in bi-directional and uni-directional mode) a new button will appear on top of the <i>Preview</i> window:  Picture 6 Note that this case represents a unidirectional communication. The Diana O. writes the information in the local database and places a request to DianaServer in order to send it to LIS computer. In this case the information related to the samples that are going to be sent is written by Diana O. instead of being downloaded from the main computer.	

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HOSTDEMO PROGRAM

Process	Expected Results
Hostdemo.exe is a technical support tool, the aim of which is to make sure that the LIS connection is properly configured. This file is not automatically installed when installing Diana System software. It can be found in Diana System software CD and shall be copied to the working directory (default:"C:\Program Files\ Grifols\ SistemaDiana"). When opening the program, it will ask for the following credentials: - User: ST - Password: d+g_diana And then the following window will appear:  Picture 1 The main window is divided into 3 sections: <i>Pending Samples</i> section is the one where the sample-test-template set requests to be transmitted to WADiana should be introduced and where the cancelled requests are displayed. <i>Processing Samples</i> section is intended to display the requests transmitted to LIS as being processed ('I' status). <i>Processed samples</i> section displays the sample-tests-template requests transmitted to LIS as completed with the corresponding results (according to the selected template). The sample-test-template sets are displayed and grouped by sample barcode (the test-template is displayed in a sublevel after unfolding the sample level). The requests can be moved from one window to another by dragging and dropping each record. The following general options are available when using HostDemo: <i>Log>></i> button displays or hides an extra window, which shows the communications and error message status. - Configuration (spanner tool) button allows configuring the communication mode (Serial Port, File or TCP/IP), which must match with the one defined in the DianaServer side. - Close (red cross) button can be used to close the program.	---

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Process	Expected Results
Several buttons can be found under each information window in each of the three sections, applying only to the corresponding section. • <i>Random</i>: Generates a random grouping of samples, tests and templates from the Diana System working directory. • <i>New</i>: Manually create a sample-test-template set/s. • <i>Edit</i>: Modify an already existing sample-test-template set. • <i>Load</i>: Allows loading a previously defined/saved worksheet from an ASTM format file with an *.hst extension. • <i>Save</i>: Save current Pending samples section requests/worksheet so that it can be loaded in an ASTM format file with extension *.hst. • <i>Delete One</i>: Deletes the selected record in the corresponding information window. • <i>Delete All</i>: Deletes the whole content of the section/window above the button.	

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LIS COMMUNICATION PROTOCOL

Process				
TEST REQUEST:				
Message header record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	H	
Delimiter Definition	Text	4	\ ^ &	
Message control ID	Null	NA		
Access Password	Null	NA		
Sender Name	Text	undefined	WADIANA^s_n^sw version	Serial number as configured in Diana System. See examples
Sender Street Address	Null	NA		
Reserved Field	Null	NA		
Sender Telephone Number	Null	NA		
Characteristics of Sender	Null	NA		
Receiver ID	Null	NA		
Comment or Special Instructions	Null	NA		
Processing ID	Text	1	P	
ASTM version	Number	1	1	
Date - time	Number	14	YYYYMMDDHHMMSS	
Request Information Record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	Q	
Sequence Number	Number	1	1	
Starting Range ID Number	Text	3	ALL	
Ending Range ID Number	Null	NA		
Universal Test ID	Text	3	ALL	
Nature of Requests Time Limits	Null	NA		
Beginning Request Results Date and Time	Null	NA		
Ending Request Results Date and Time	Null	NA		
Requesting Physician Name	Null	NA		
Requesting Physician Telephone Number	Null	NA		
User Field 1	Null	NA		
User Field 2	Null	NA		
Request Information Status Codes	Text	1	I	

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Process				
Message terminator record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	L	
Sequence Number	Number	1	1	
Termination Code	Text	1	N	
Examples: WADiana: H \^& WADIANA^0100^5 P 1 20110603164236 Q 1 ALL ALL L 1 N				
<u>RESPONSE TO THE REQUEST:</u>				
Message header record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	H	
Delimiter Definition	Text	4	\ ^&	
Message control ID	Null	NA		Message Record can finish in the line above as no further information is processed.
Access Password	Null	NA		
Sender Name	Null	NA	WADIANA^s_n^sw version	This data is not required, but may be present in the message. If present, it must be coincident with the one corresponding to the instrument requiring the information.
Sender Street Address	Null	NA		
Reserved Field	Null	NA		
Sender Telephone Number	Null	NA		
Characteristics of Sender	Null	NA		
Receiver ID	Null	NA		
Comment or Special Instructions	Null	NA		
Processing ID	Null	NA		
ASTM version	Null	NA		
Date - time	Null	NA		

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Process				
Patient information record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	P	
Sequence Number	Number	undefined	Number within LIS message	
Practice-Assigned Patient ID	Null	undefined	Patient ID	Patient ID must be populated with a value. If the LIS system is not using this field, Sample BC is recommended.
Laboratory-Assigned Patient ID	Null	undefined		
Patient ID Number 3	Null	undefined		
Patient Name	Null	100	Last name^First name^Middle name or initial^Suffix^Title	
Mother's Maiden Name	Null	undefined		
Birthdate	Date/Time	8	YYYYMMDD	
Patient Sex	Text	1	M,F or U	
Patient Race-Ethnic Origin	Null	undefined		Message can finish in the line above as no further information is processed.
Patient Address	Null	undefined		In case no demographic data sent or partial demographic data, message record can finish after <i>Patient ID</i> field or after last demographic data sent.
Reserved Field	Null	undefined		
Patient Telephone Number	Null	undefined		
Attending Physician ID	Null	undefined		
Special Field 1	Null	undefined		
Special Field 2	Null	undefined		
Patient Height	Null	undefined		
Patient Weight	Null	undefined		
Patient's Known or Suspected Diagnosis	Null	undefined		
Patient Active Medications	Null	undefined		
Patient's Diet	Null	undefined		
Practice Field Number 1	Null	undefined		
Practice Field Number 2	Null	undefined		
Admission and Discharge Dates	Null	undefined		
Admission Status	Null	undefined		
Nature of Alternative Diagnostic Code and Classifiers	Null	undefined		
Alternative Diagnostic Code and Classification	Null	undefined		
Patient Religion	Null	undefined		
Marital Status	Null	undefined		
Isolation Status	Null	undefined		
Language	Null	undefined		
Hospital Service	Null	undefined		
Hospital Institution	Null	undefined		
Dosage Category	Null	undefined		

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Process				
Test order record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	O	
Sequence Number	Number	undefined	Number within Patient record	
Specimen ID	Text	undefined	Sample BC (non Xmatch) Sample BC^Donor BC (Xmatch)	
Instrument Specimen ID	Null	undefined		
Universal test ID	Text	undefined	^^Test name^Template name	In order to complete bidirectional connection, template field must be sent from LIS.
Priority	Text	1		Priority is ignored. All orders are considered as routine. STAT orders cannot be sent from LIS. Message can finish in the line above as no further information is processed.
Requested/Ordered Date and Time	Null	undefined		
Specimen Collection Date and Time	Null	undefined		
Collection End Time	Null	undefined		
Collection Volume	Null	undefined		
Collector ID	Null	undefined		
Action Code	Null	undefined		
Danger Code	Null	undefined		
Relevant Clinical Information	Null	undefined		
Date/Time Specimen Received	Null	undefined		
Specimen Descriptor	Null	undefined		
Ordering Physician	Null	undefined		
Physician's Telephone Number	Null	undefined		
User Field Number 1	Null	undefined		
User Field Number 2	Null	undefined		
Laboratory Field Number 1	Null	undefined		
Laboratory Field Number 2	Null	undefined		
Date/Time Results Reported or Last Modified	Null	undefined		
Instrument Charge to Information System	Null	undefined		
Instrument Section ID	Null	undefined		
Report Types	Null	undefined		
Reserved Field	Null	undefined		
Location of Specimen Collection	Null	undefined		
Nosocomial Infection Flag	Null	undefined		
Specimen Service	Null	undefined		
Specimen Institution	Null	undefined		

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Process				
Message terminator record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	L	
Sequence Number	Number	1	1	
Termination Code	Text	1	N	
Example: H \^& P 1 1001 2001 Apell1^Nom1 19751201 U O 1 01 ^D^DGW01A.gru^Grupo.pln P 2 1004 2004 Apell4^Nom4 19751204 U O 1 04 ^D^DGW01A.gru^Grupo.pln P 3 1005 2005 Apell5^Nom5 19751205 U O 1 05 ^D^DGW01A.gru^Grupo.pln P 4 1006 2006 Apell6^Nom6 19751206 U O 1 06 ^D^DGW01A.gru^Grupo.pln L 1 N Note: character ‘&’ sent as a part of sample identification of demographic data will cause communication errors.				
<u>CANCELLED-IN PROCESS SAMPLE:</u>				
Message header record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	H	
Delimiter Definition	Text	4	\ ^&	
Message control ID	Null	NA		
Access Password	Null	NA		
Sender Name	Text	undefined	WADIANA^s_n^sw version	Serial number as configured in Diana System. See examples
Sender Street Address	Null	NA		
Reserved Field	Null	NA		
Sender Telephone Number	Null	NA		
Characteristics of Sender	Null	NA		
Receiver ID	Null	NA		
Comment or Special Instructions	Null	NA		
Processing ID	Text	1	P	
ASTM version	Number	1	1	
Date - time	Number	14	YYYYMMDDHHMMSS	

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Process				
Patient information record:				
ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	P	
Sequence Number	Number	undefined	Number within LIS message	
Practice-Assigned Patient ID	Null	undefined	Patient ID	
Laboratory-Assigned Patient ID	Null	undefined		
Patient ID Number 3	Null	undefined		
Patient Name	Null	100	Last name^First name^Middle name or initial^Suffix^Title	WADiana does not interpret this information and does not separate Last name from First name, etc...
Mother's Maiden Name	Null	undefined		
Birthdate	Date/Time	8	YYYYMMDD	
Patient Sex	Text	1	M,F or U	
Patient Race-Ethnic Origin	Null	undefined		Message Record ends in the line above. In case no demographic data sent or partial demographic data, message record finishes after <i>Patient ID</i> field or after last demographic data sent
Patient Address	Null	undefined		
Reserved Field	Null	undefined		
Patient Telephone Number	Null	undefined		
Attending Physician ID	Null	undefined		
Special Field 1	Null	undefined		
Special Field 2	Null	undefined		
Patient Height	Null	undefined		
Patient Weight	Null	undefined		
Patient's Known or Suspected Diagnosis	Null	undefined		
Patient Active Medications	Null	undefined		
Patient's Diet	Null	undefined		
Practice Field Number 1	Null	undefined		
Practice Field Number 2	Null	undefined		
Admission and Discharge Dates	Null	undefined		
Admission Status	Null	undefined		
Nature of Alternative Diagnostic Code and Classifiers	Null	undefined		
Alternative Diagnostic Code and Classification	Null	undefined		
Patient Religion	Null	undefined		
Marital Status	Null	undefined		
Isolation Status	Null	undefined		
Language	Null	undefined		
Hospital Service	Null	undefined		
Hospital Institution	Null	undefined		
Dosage Category	Null	undefined		

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Test order record:

ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	O	
Sequence Number	Number	undefined	Number within Patient record	
Specimen ID	Text	undefined	Sample BC (non Xmatch) Sample BC^Donor BC (Xmatch)	
Instrument Specimen ID	Null	undefined	In case of In Process samples: Batch number-Position in carousel	See examples
Universal test ID	Text	undefined	^^^Test name^Template name	
Priority	Text	1	R	R for orders executed as routine. STAT samples cannot be sent from LIS
Requested/Ordered Date and Time	Null	undefined	YYYYMMDDHHMMSS	Same as timestamp as in H record
Specimen Collection Date and Time	Null	undefined		
Collection End Time	Null	undefined		
Collection Volume	Null	undefined		
Collector ID	Null	undefined		
Action Code	Null	undefined		
Danger Code	Null	undefined		
Relevant Clinical Information	Null	undefined		
Date/Time Specimen Received	Null	undefined		
Specimen Descriptor	Null	undefined		
Ordering Physician	Null	undefined		
Physician's Telephone Number	Null	undefined		
User Field Number 1	Null	undefined		
User Field Number 2	Null	undefined		
Laboratory Field Number 1	Null	undefined		
Laboratory Field Number 2	Null	undefined		
Date/Time Results Reported or Last Modified	Null	undefined		
Instrument Charge to Information System	Null	undefined		
Instrument Section ID	Null	undefined	WADIANA^s_n^sw version	
Report Types	Text	1	X for Cancelled Samples I for In Process Samples	
Reserved Field	Null	undefined		Message Record ends in the line above
Location of Specimen Collection	Null	undefined		
Nosocomial Infection Flag	Null	undefined		
Specimen Service	Null	undefined		
Specimen Institution	Null	undefined		

Message terminator record:

ASTM Field	Type	Length	WADiana	Comments
Record type ID	Text	1	L	
Sequence Number	Number	1	1	
Termination Code	Text	1	N	

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Process

Example:

```
H|\^&||WADIANA^0100^5|||||P|1|20111019171022
P|1|1001|||Apell1^Nom1||19751201|U|
O|1|01|0108-
1|^^^DGW01A.gru^Grupo.pln|R|20111019171022|||||||||||||WADIANA^0100^5||
P|2|1004|||Apell4^Nom4||19751204|U|
O|1|04|0108-
2|^^^DGW01A.gru^Grupo.pln|R|20111019171022|||||||||||||WADIANA^0100^5||
P|3|1005|||Apell5^Nom5||19751205|U|
O|1|05|^^^DGW01A.gru^Grupo.pln|R|20111019171022|||||||||||||WADIANA^0100^5|X|
P|4|1006|||Apell6^Nom6||19751206|U|
O|1|06|^^^DGW01A.gru^Grupo.pln|R|20111019171022|||||||||||||WADIANA^0100^5|X|
L|1|N
```

RESULTS:

Message header record:

ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments
Record type ID	Text	1	H	H	
Delimiter Definition	Text	4	\ ^&	\ ^&	
Message control ID	Null	NA			
Access Password	Null	NA			
Sender Name	Text	undefined	WADIANA^s_n^sw version	REPORT^000^Diana O.	Serial number as configured in Diana System. See examples
Sender Street Address	Null	NA			
Reserved Field	Null	NA			
Sender Telephone Number	Null	NA			
Characteristics of Sender	Null	NA			
Receiver ID	Null	NA			
Comment or Special Instructions	Null	NA			
Processing ID	Text	1	P	P	
ASTM version	Number	1	1	1	
Date - time	Number	14	YYYYMMDDHHMMSS	YYYYMMDDHHMMSS	

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Class:	Control del Diseño		Effective date:	24-Apr-2018
Title:	SM DG57. LIS connection			

Process					
Patient information record:					
ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments
Record type ID	Text	1	P	P	
Sequence Number	Number	undefined	Number within LIS message	Number within LIS message	
Practice-Assigned Patient ID	Null	undefined	Patient ID	Sample BC	
Laboratory-Assigned Patient ID	Null	undefined			Message Record ends in the line above for uni-directional case
Patient ID Number 3	Null	undefined			
Patient Name	Null	100	Last name^First name^Middle name or initial^Suffix^Title		Exactly same as sent from LIS
Mother's Maiden Name	Null	undefined			
Birthdate	Date/Time	8	YYYYMMDD		
Patient Sex	Text	1	M,F or U		
Patient Race-Ethnic Origin	Null	undefined			Message Record in the line above for bi-directional case. In case no demographic data sent or partial demographic data, message record finishes after <i>Patient ID</i> field or after last demographic data sent
Patient Address	Null	undefined			
Reserved Field	Null	undefined			
Patient Telephone Number	Null	undefined			
Attending Physician ID	Null	undefined			
Special Field 1	Null	undefined			
Special Field 2	Null	undefined			
Patient Height	Null	undefined			
Patient Weight	Null	undefined			
Patient's Known or Suspected Diagnosis	Null	undefined			
Patient Active Medications	Null	undefined			
Patient's Diet	Null	undefined			
Practice Field Number 1	Null	undefined			
Practice Field Number 2	Null	undefined			
Admission and Discharge Dates	Null	undefined			
Admission Status	Null	undefined			
Nature of Alternative Diagnostic Code and Classifiers	Null	undefined			
Alternative Diagnostic Code and Classification	Null	undefined			
Patient Religion	Null	undefined			
Marital Status	Null	undefined			
Isolation Status	Null	undefined			
Language	Null	undefined			
Hospital Service	Null	undefined			

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Process					
ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments
Hospital Institution	Null	undefined			
Dosage Category	Null	undefined			
Test order record:					
ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments
Record type ID	Text	1	O	O	
Sequence Number	Number	undefined	Number within Patient record	Number within Patient record	
Specimen ID	Text	undefined	Sample BC (non Xmatch) Sample BC^Donor BC (Xmatch)	Sample BC (non Xmatch) Sample BC^Donor BC (Xmatch)	
Instrument Specimen ID	Text	7	Batch number-Position of sample in the carousel (xxxx-yy)		
Universal test ID	Text	undefined	^^^Test name^Template name	^^^Template name	
Priority	Text	1	R	R	
Requested/Ordered Date and Time	Null	undefined	YYYYMMDDHHMMSS	YYYYMMDDHHMMSS	Same as in <i>In Process/Cancel</i> Message
Specimen Collection Date and Time	Null	undefined			
Collection End Time	Null	undefined			
Collection Volume	Null	undefined			
Collector ID	Null	undefined			
Action Code	Null	undefined			
Danger Code	Null	undefined			
Relevant Clinical Information	Null	undefined	#Batch number^BCs of Reagents and cards used, separated by '^'	#Batch number^BCs of Reagents and cards used, separated by '^'	This option is user configurable (see note below). If sample results corresponds to different batches (may happen in unidirectional mode), reagent BCs are grouped under each batch number: See examples.
Date/Time Specimen Received	Null	undefined			
Specimen Descriptor	Null	undefined			
Ordering Physician	Null	undefined			
Physician's Telephone Number	Null	undefined			
User Field Number 1	Null	undefined			
User Field Number 2	Null	undefined			
Laboratory Field Number 1	Null	undefined			
Laboratory Field Number 2	Null	undefined			
Date/Time Results Reported or Last Modified	Null	undefined			
Instrument Charge to Information System	Null	undefined			
Instrument Section ID	Null	undefined	WADIANA^s_n^sw version	REPORT^000^Diana O.	
Report Types	Text	1	F	F	
Reserved Field	Null	undefined			Message Record ends at the line above

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Process																																																																																																																							
ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments																																																																																																																		
Location of Specimen Collection	Null	undefined																																																																																																																					
Nosocomial Infection Flag	Null	undefined																																																																																																																					
Specimen Service	Null	undefined																																																																																																																					
Specimen Institution	Null	undefined																																																																																																																					
Note: The transmission to LIS of cards and reagents barcodes used to process each sample can be configured by means of Windows registry HKEY_LOCAL_MACHINE\SOFTWARE\Grifols\SistemaDiana\Diana O.\SendBCReacSupp: SendBCReacSupp = 0: Barcodes are not transmitted (default value). SendBCReacSupp = 1: Barcodes are transmitted. Result record: <table> <tr> <th>ASTM Field</th><th>Type</th><th>Length</th><th>Bidirectional</th><th>Unidirectional</th><th>Comments</th></tr> <tr> <td>Record type ID</td><td>Text</td><td>1</td><td>R</td><td>R</td><td></td></tr> <tr> <td>Sequence Number</td><td>Number</td><td>undefined</td><td>Number within Order Record</td><td>Number within Order Record</td><td></td></tr> <tr> <td>Universal test ID</td><td>Null</td><td>undefined</td><td>^^^Test name^Template name</td><td>^^^Template name</td><td></td></tr> <tr> <td>Data or Measurement Value</td><td>Text</td><td>undefined</td><td>Results included in template, separated by '^'</td><td>Results included in template, separated by '^'</td><td></td></tr> <tr> <td>Units</td><td>Text</td><td>undefined</td><td>Template fields, in the same order of the results, separated by '^'</td><td>Template fields, in the same order of the results, separated by '^'</td><td></td></tr> <tr> <td>Reference Ranges</td><td>Null</td><td>undefined</td><td></td><td></td><td></td></tr> <tr> <td>Result Abnormal Flags</td><td>Null</td><td>undefined</td><td>N</td><td>N</td><td></td></tr> <tr> <td>Nature of Abnormality Testing</td><td>Null</td><td>undefined</td><td></td><td></td><td></td></tr> <tr> <td>Result Status</td><td>Null</td><td>undefined</td><td>F</td><td>F</td><td></td></tr> <tr> <td>Date of Change in Instrument Normative Values or Units</td><td>Null</td><td>undefined</td><td></td><td></td><td></td></tr> <tr> <td>Operator Identification</td><td>Text</td><td>undefined</td><td>Username of sender</td><td>Username of sender</td><td></td></tr> <tr> <td>Date/Time Test Started</td><td>Date/Time</td><td>undefined</td><td>YYYYMMDDHHMMSS</td><td>YYYYMMDDHHMMSS</td><td>Same as in O Record</td></tr> <tr> <td>Date/Time Test Completed</td><td>Date/Time</td><td>undefined</td><td>YYYYMMDDHHMMSS</td><td>YYYYMMDDHHMMSS</td><td></td></tr> <tr> <td>Instrument Identification</td><td>Null</td><td>undefined</td><td>WADIANA^s_n^sw version</td><td>REPORT^000^Diana O.</td><td></td></tr> </table> Message terminator record: <table> <tr> <th>ASTM Field</th><th>Type</th><th>Length</th><th>Bidirectional</th><th>Unidirectional</th><th>Comments</th></tr> <tr> <td>Record type ID</td><td>Text</td><td>1</td><td></td><td>L</td><td></td></tr> <tr> <td>Sequence Number</td><td>Number</td><td>1</td><td></td><td>1</td><td></td></tr> <tr> <td>Termination Code</td><td>Text</td><td>1</td><td></td><td>N</td><td></td></tr> </table>						ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments	Record type ID	Text	1	R	R		Sequence Number	Number	undefined	Number within Order Record	Number within Order Record		Universal test ID	Null	undefined	^^^Test name^Template name	^^^Template name		Data or Measurement Value	Text	undefined	Results included in template, separated by '^'	Results included in template, separated by '^'		Units	Text	undefined	Template fields, in the same order of the results, separated by '^'	Template fields, in the same order of the results, separated by '^'		Reference Ranges	Null	undefined				Result Abnormal Flags	Null	undefined	N	N		Nature of Abnormality Testing	Null	undefined				Result Status	Null	undefined	F	F		Date of Change in Instrument Normative Values or Units	Null	undefined				Operator Identification	Text	undefined	Username of sender	Username of sender		Date/Time Test Started	Date/Time	undefined	YYYYMMDDHHMMSS	YYYYMMDDHHMMSS	Same as in O Record	Date/Time Test Completed	Date/Time	undefined	YYYYMMDDHHMMSS	YYYYMMDDHHMMSS		Instrument Identification	Null	undefined	WADIANA^s_n^sw version	REPORT^000^Diana O.		ASTM Field	Type	Length	Bidirectional	Unidirectional	Comments	Record type ID	Text	1		L		Sequence Number	Number	1		1		Termination Code	Text	1		N	
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Process

LIS ERRORS.

In this section WADiana and LIS communication errors are listed. Following tables include Diana System errors related to LIS communication.

Module	Error code	Cause
Message contents errors (ASTM-1394 standard)	Host format error #1	Heading error: wrong separators.
	Host format error #2	Heading error: first line of the message does not begin with 'H'.
	Host format error #3	Error Reading Patient No patient ID.
	Host format error #4	Test: Not enough fields.
	Host format error #5	Test: No specimen field.
	Host format error #6	Test: No test name fields.
	Host format error #7	Error, an O field before a P field.
	Host format error #8	Error, an R field before an O field.
	Host format error #9	Results: Not enough fields.
	Host format error #10	File received has no lines, it is empty.
File communication errors	Host file error #1	Writing error in the exchange file.
	Host file error #2	Writing error in the exchange file.
	Host file error #3	Exchange file cannot be deleted.
	Host file error #4	The exchange file still exists after concluding the Transmission.
TCP/IP communication errors	Remote host not ready	LIS server is not ready when trying to establish communication with DianaServer.
Serial or TCP/IP communication errors	Host serial error #1	Time-out when receiving frame.
	Host serial error #2	Incorrect frame received.
	Host serial error #3	Incorrect frame received, STX fault.
	Host serial error #4	Neither ETB or ETX being received in the frame.
	Host serial error #5	Frame received without checksum.
	Host serial error #6	Frame received with wrong checksum.
	Host serial error #7	Frame received with wrong ID.
	Host serial error #8	Communication error in establishing the connection while awaiting reception.
	Host serial error #9	Undetermined error.
	Host serial error #10	Frame sent 6 times and returned by LIS as faulty (NAK).
	Host serial error #11	ENQ received while establishing communication.
	Host serial error #12	Strange character received while establishing communication.
	Host serial error #13	EOT missing.

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	Document REGD-0000782_v3 replaces document 57009VOM001.	Document reference changed.
4.0	2011/07/19	Added previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
4.0	2011/07/19	Document REGD-0000782_v4 replaces document REGD-0000782_v3.	Document version changed.
4.0	2011/07/19	Pictures updated.	Updated pictures to new software version.
4.0	2011/07/19	Document title.	Commercial reference substituted by project reference.
4.0	2011/07/19	Document header.	Increase information to easy document search.
5.0	2012/10/29	Include procedure 2 explaining how to use more than one DianaServer software in the same network domain.	Work with more than one DianaServer in the same network domain.
6.0	October 2013	Document REGD-0000782_v5 replaces document REGD-0000782_v4.	Document version changed.
6.0	October 2013	'LIS connection' title replaces 'connection to host'.	Standardize title with other projects.
6.0	October 2013	Added following explanations: -LIS connection. -HostDemo software. -LIS communication protocol.	LIS Information increased.
7.0	December 2014	Procedure description	Procedure 2 applies to Windows XP OS
7.0	December 2014	The necessary Windows registry configuration to enable/disable the transmission to LIS of cards and reagents barcodes is detailed.	LIS information increased.
8.0	April 2016	Diana Server configuration note is added.	Diana Server configuration details increased.
9.0	April 2018	Procedure description (title page)	Improve procedure description accuracy
9.0	April 2018	Updated <i>LIS connection</i> section description	New TCP/IP mode
9.0	April 2018	<i>Procedure 2</i> has been deleted.	New DianaServer characteristics
9.0	April 2018	<i>Procedure 1</i> renamed to <i>Procedure</i> .	Only one procedure is described
9.0	April 2018	Updated <i>HostDemo</i> section description	New TCP/IP mode
9.0	April 2018	<i>LIS communication protocol</i> section: new values for protocol fields, updated comments and removed <i>Field length limitations</i> section.	<i>Instrument section ID</i> field was not listed as an ASTM field. Host.db database is not used, so there are no more field length limitations.
9.0	April 2018	Updated <i>LIS errors</i> section	New TCP/IP mode

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